Note on Chiral Topological Order on Lattice

X. D. H.

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Abstract

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1 Lattice Chiral TO from VOA

1.1 Lightning Review of VOA

A physicist's quick guide: VOA = chiral half of a 2D CFT, packaged algebraically.

1.1.1 The Core Idea

In a 2D CFT, every local field splits into chiral (z-dependent) and anti-chiral ($\overline{z}$-dependent) parts. Remember in RCFT the fields can be decompose as:

$$\phi_i(z, \overline{z}) = \sum_{a,b} C_i^{a\overline{b}} \Phi_a(z) \otimes \overline{\Phi}_{\overline{b}}(\overline{z}) \tag{1}$$

These chiral fields $a(z), b(z), ...$ close under OPE:

$$a(z)b(w) \sim \sum_n \frac{c_n(w)}{(z-w)^n} \tag{2}$$

A **vertex operator algebra** (VOA) is just the rigorous packaging of "all chiral fields + their OPEs" as one algebraic object $\mathcal{V}$.

1.1.2 The Data of a VOA

A VOA is a tuple $(V, |0\rangle, T, Y, \omega)$:

Math object	Physics meaning
vector space V	Hilbert space of chiral states (state-operator correspondence)
vacuum $\|0\rangle \in V$	vacuum state $\leftrightarrow$ identity field $\mathbb{1}$
translation $T = L_{-1}$	generator of ∂_z
vertex map $Y(a, z) : V \to \mathrm{End}(V)$	sends state a to its field $a(z)$
conformal vector ω	stress tensor $T(z) = Y(\omega, z)$, central charge c

Everything is determined by **one core relation** — locality:

$$(z - w)^N [Y(a, z), Y(b, w)] = 0 \quad \text{for some} \quad N \geq 0 \tag{3}$$

This is just the statement that the OPE has finitely many singular terms.

1.1.3 Modules = Primary Sectors

A **module** M of $\mathcal{V}$ is a vector space carrying a consistent action of all chiral fields in $\mathcal{V}$. Concretely, a module is specified by the following data:

Data	Physical meaning
vector space M	Hilbert space of one primary sector
action map $Y_M(a, z) : \mathcal{V} \to \mathrm{End}(M)$	every chiral field $a(z)$ acts on this sector
L_0-grading $M = \bigoplus_{n \geq 0} M_{h+n}$	primary at weight h + tower of descendants at level n
lowest-weight state $\|h\rangle \in M_h$	primary state: $L_0 \ \|h\rangle = h \ \|h\rangle$, $L_{n>0}\|h\rangle = 0$
conformal weight h (lowest L_0 eigenvalue)	scaling dimension of the primary field

The action Y_M must satisfy the same locality / Jacobi axioms as the VOA's own vertex map Y — this guarantees that the OPE structure of $\mathcal{V}$ is preserved on M.

one irreducible module $\leftrightarrow$ one primary field + all its descendants

Note. All descendants in M are reached from $|h\rangle$ by acting with negative-mode operators ($L_{-n}, J_{-n}, \psi_{-r}$, etc., for $n, r > 0$). The module is **irreducible** if it cannot be decomposed into smaller submodules — equivalently, if it corresponds to a single primary.

1.1.3.1 The Vacuum Module

The most important example: $\mathcal{V}$ is always a module over itself, called the **vacuum module**. Its lowest-weight state is the vacuum $|0\rangle$ with $h = 0$, and its descendants

$$|0\rangle, \quad L_{-2}|0\rangle, \quad J_{-1}|0\rangle, \quad \psi_{-\frac{1}{2}}|0\rangle, \quad ... \tag{4}$$

correspond (via state-operator correspondence) to **all chiral fields of the CFT**: $\mathbb{1}, T(z), J(z), \psi(z), : T^2 : (z), \partial T(z), ...$

In other words: *the vacuum module is the VOA itself.* It contains the identity field, the stress tensor, and all chiral generators. In $\mathrm{Rep}(\mathcal{V})$ it is the **unit object** (labeled $a = 1$), and acts trivially under fusion: $V \otimes V^{(a)} = V^{(a)}$. Other irreducible modules $V^{(a)}$ ($a \neq 1$) are *distinct* sectors with primary weight $h_a > 0$ — they correspond to nontrivial anyons.

Examples. Ising VOA: three modules $\mathbb{1}, \psi, \sigma$ with $h = 0, \frac{1}{2}, \frac{1}{16}$. Compact boson $V_{U(1)_m}$: m modules labeled by $p = 0, ..., m - 1$ with $h_p = \frac{p^2}{2}m$. In both cases, $p = 0$ (resp. $\mathbb{1}$) is the vacuum module — the VOA itself.

1.1.4 The Three Modifiers

For a VOA to describe a sensible RCFT, three conditions matter:

- **Unitary**: V has positive-definite inner product, $L_n^\dagger = L_{-n}$. (Real physical CFT.)
- **Rational**: only *finitely many* irreducible modules. (Finitely many primaries, finitely many anyons.)
- **Good**: mild technical conditions (e.g. C_2-cofiniteness) ensuring $\mathrm{Rep}(\mathcal{V})$ is a modular tensor category.

A **good unitary rational VOA** is exactly the chiral data of an RCFT with a well-defined MTC structure.

1.2 Sopenko's Construction

Bibliography